Chapter – 10: Life Processes in Plants

- Grade 6 – learnt – all living beings – grow – need food – their growth
- Previous chapter – processes – animals – obtain nutrition
- Animals – need food – BUT – plants – do they eat?
- Animals grow – size, weight – increase – bodies – various changes – what changes – obs. in plants?
- Learnt – food – nutrients – carbs, fats, proteins, minerals, vitamins, water – essential – growth
- This chapter – **explore** – plants – obtain nutrients – their growth

How do Plants Grow?

- Look around – neighbourhood – obs. changes – plants – its life span
- Plant grows – new leaves, branches – height increases – stem thickens
- Discuss with friends – provide explanation
    - Water plants – grow better – contribute – growth
    - Maybe – take food – soil – roots
    - Sunlight – plays some role – growth
- Activity –
    - Take – 3 earthen pots – same size – fill – garden soil – plant saplings – similar size – fast-growing plant – chilli, tomato – each pot
    - Label pots – A, B, C
    - Count – no. of leaves – each plant – record
    - Place – pot A – direct sunlight – keep soil – moist – add water – daily
    - Place – pot B – direct sunlight – no water
    - Place – pot C – dark – keep soil – moist – add water – daily
    - **Obs.** plants – 2 weeks – record changes – height, no. of leaves, other changes – **record**
- **Analyse** – obs. – discuss – teacher, friends
- Find – pot A – grows better than – pot C
- Pot B – may die – Sunlight – BUT – no water
- Results – indicate – plants req. – both – sunlight, water – growth

How do Plants get Food for Their Growth?

- Animals – food – plants, other animals – BUT – how – plants – obtain nutrition?
- Like animals – plants – do not *eat* food

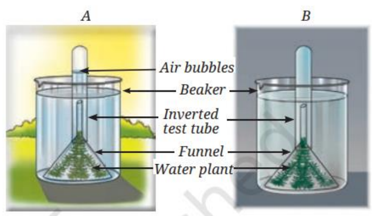
Leaves: the 'food factories' of plants

- Plants – store food – starch – carbohydrate
- This starch – produced in leaves – design – broad, flat
- Leaves – mostly green – presence – green pigment – **chlorophyll** – helps – capture sunlight
- Activity –
    - Keep – leaf – boiling water – 5 min. – soften it
    - Dip – leaf – test tube – alcohol
    - Place test tube – beaker – boiling water – wait – leaf – colourless
    - Take out leaf – place it – plate
    - Put – few drops – dil. iodine solu. – dropper – decolorized leaf – wait – few min. – obs.

  - Colour – leaf – blue-black – indicate – presence – starch
- Above act. – potted plant – learnt – water, sunlight – essential – plant growth
- Above act. – iodine solu. – green leaves – store starch – food
- Bhaskar – loves gardening – free time – very curious – look around – wonder – plants – produce food
- His experience – Bhaskar knows – water, sunlight – essential – plant growth
- BUT – wonders – sunlight – contribute – food production
- Activity –
  - Bhaskar – look leaf – green, non-green patches – 2 potted plants – one – sunlight – one dark – 36 hrs.
  - Compare leaves – before, after – starch test
  - Made sketch – leaves – record – location – green, non-green patches – help – tracing paper
  - After that – iodine test – recorded obs.
- Bhaskar – recorded – blue-black colour – green patches – plant – sunlight
- Also recorded – no change in colour – green patches – plant – dark – indicate – no starch produced
- Non-green patches – not sufficient chlorophyll – enough starch – detected – iodine test
- Above activity – leaves – mostly green – presence – chlorophyll
- Also seen – starch produced – green patches – present
- We infer – chlorophyll – helps – prepare starch – sunlight presence
- Sunlight – essential – starch preparation – hence – leaves – food factories

Role of air in preparation of food

- Activity –
  - Take – potted plant – keep in dark – 2-3 days – locate 1 leaf – this plant
  - Take – wide-mouth bottle – pour – caustic soda (NaOH) – absorb – CO_2
  - Insert – half leaf – inside bottle – split cork – leave other half – outside
  - Place setup – sunlight – few hrs.
  - Obs. – record – availability – water, sunlight, chlorophyll, CO_2
  - Remove leaf – iodine test – record obs.
- **Notice** – part of leaf – outside bottle – blue-black – indicate – presence of starch
- Part of leaf – inside bottle – no colour change – indicate – no food production
- Reason – caustic soda – absorb – CO_2 – this exp. – CO_2 – essential – prepare starch
- Learnings so far – **sunlight**, **water**, **chlorophyll**, **CO_2** – essential – food synthesis
- This process – plants – prepare food – presence of sunlight, chlorophyll – **photosynthesis**
- Leaf – primary site – photosynthesis – other parts – contain chlorophyll – perform photosynthesis
- This process – plants – take many things – BUT – do they also release something?
- Activity –
  - Look – fig. – **compare** – 2 setups – A, B – analyse
  - This fig. – setup A – sunlight – setup B – dark
  - Obs. – setup A – gas bubbles – emerge, accumulate – inverted test tube



- Children – discuss – results –
  - 1st – I remember – science lab – similar setup
  - 2nd – sufficient gas – accumulated – inverted tube – tube off – mouth covered – introduced – lit matchstick – intense flame
  - 1st – didi inferred – gas – test tube – rich in O_2 – indicate – O_2 release – photosynthesis – also indicate – photosynthesis – req. sunlight

Photosynthesis: in a nutshell

- We know – water, sunlight, CO_2, chlorophyll – necessary – photosynthesis – produce carbs
- This process – food produced – glucose – simple carb
- This glucose – instant src. – energy – later – converted to starch
- Word equation – photosynthesis –
    - Carbon dioxide + Water $\xrightarrow[\text{Chlorophyll}]{\text{Sunlight}}$ Glucose + Oxygen

How do leaves exchange gases during photosynthesis?

- We know – photosynthesis req. CO_2 – release O_2
- **Conduct** – act. – understand – exchange of gases
- Activity –
    - Collect – leaf – rhoeo, money plant, onion, hibiscus, coleus, grass
    - Put in beaker – fill water
    - Carefully – peel – thin layer – lower surface – leaf
    - Place peel – watch glass – add water
    - Take – microscope slide – put – water drop
    - Use forceps – transfer peel – watch glass to slide
    - Put – drop of ink – leaf peel – dropper
    - Cover peel – coverslip – obs. – under microscope
- Notice – tiny pores – **stomata** – surface of leaves – help – exchange of gases

Transport in plants

Transport of water and minerals

- All beings – need water – plant – use water – photosynthesis
- Water – with minerals – taken up – through roots – minerals – imp. nutrients
- How – water, minerals – move – all parts?
- Study – water transport – act. – req. – 2 glass tumblers, some water, red ink, twigs of 2 tender plants with white flowers – white *sadabahar*, balsam
- Activity –
    - Take – 2 tumblers – label – A, B
    - Each tumbler – fill 1/3rd – water
    - Add – few drops – red ink – tumbler B
    - Cut stems – both plants – their base – oblique – keep inside water – place – 1 in each tumbler
    - Obs. – next few days
- Compare – plant stems – both tumbler
- Obs. – red colour – stem, leaves, flowers – plant – tumbler B
- Cut stem – upper part – tumbler B – obs. – magnifying glass
- Spot – red colour – reason – thin tube-like structure – **xylem** – all parts
- Similarly – minerals – dissolved in water – move up – stem
- We know – transport – water, minerals – all parts – xylem
- How – food – transport?

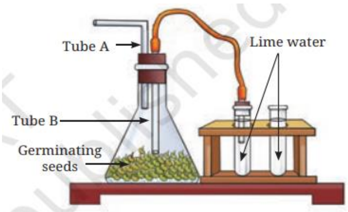
Transport of food

- We know – leaves – primary site – photosynthesis
- Food prepared – leaves – transport – all parts – another set – thin tube structures – **phloem**

- Transported food – stored – other parts – seeds, roots

Do Plants Respire?

- Grade 6 – *curiosity* – chapter – living creatures – learnt – all living beings – respire
- Do plants – respire – like us?
- Activity –



  - Soak – *moong* seeds – water – overnight
  - Put – cotton layer – conical flask – moisten cotton – water
  - Place – soaked seeds – wet cotton – flask
  - Cover mouth – conical flask – cork – 2 holes
  - Fit – 2 tubes – A, B – 2 holes – leave it – undisturbed – 24 hrs. – dark
  - Take – 2 test tubes – fill – lime water
  - Cover mouth – 1 test tube – cork – 1 hole
  - Dip – 1 glass tube – hole – cork – test tube
  - Connect flask – test tube – rubber pipe – as shown
- Compare – both test tubes – colour change
- Test tube – connected – pipe – milky – indicate – presence – more CO_2
- Air – small qty. – CO_2 – naturally
- Flask – additional CO_2 – seeds respire
- Respiration – glucose – break down – presence of O_2 – release CO_2, water, energy
- Word eq. – respiration –
  - Glucose + Oxygen → Carbon dioxide + water + energy
- Energy produced – respiration – plants use – growth, development
- All part – green or not – respire
- Plants – diff. mechanisms – synthesis, transport, utilise food – get energy